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Unit 2 - Logical operators

PDF Documents

PDF Document 1: 1734182998763-Chapter 3_unit2 Logical operators.pdf

This PDF document is included in the unit materials.

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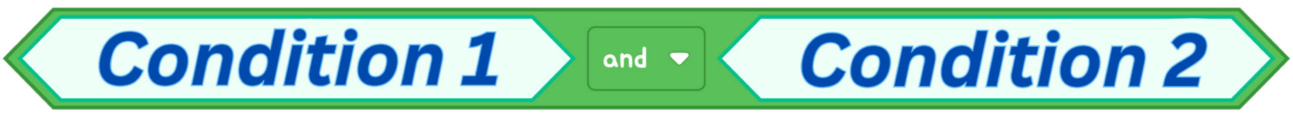
UNIT 2 LOGICAL OPERATORS

AND

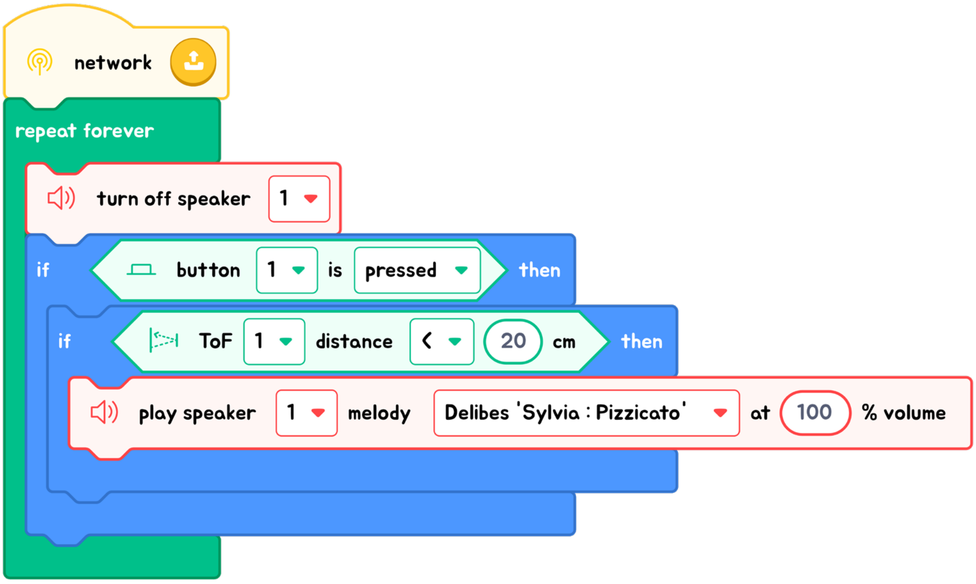
Imagine you and your friend want to play outside, but both of you have homework to finish. You agree that you can only go out if both of you complete your homework. This agreement is like using the "AND" operator in programming.

If you say, "I will finish my homework AND you will finish yours," it means both conditions must be true for you to go outside. If either of you doesn't finish your homework, you both stay inside.

In programming, when you use the "AND" operator in an if statement, it ensures that both conditions must be true for the code block to execute. So, if your if statement includes two conditions separated by "AND," both must be satisfied for the code within the if statement to run.



EXPLANATION WITH CODE EXAMPLE



In the code example above, we used nested if statements to check two conditions for one task. Below, we demonstrate how to transform these nested ifs into a single if statement using the 'and' operator.

Code Insight

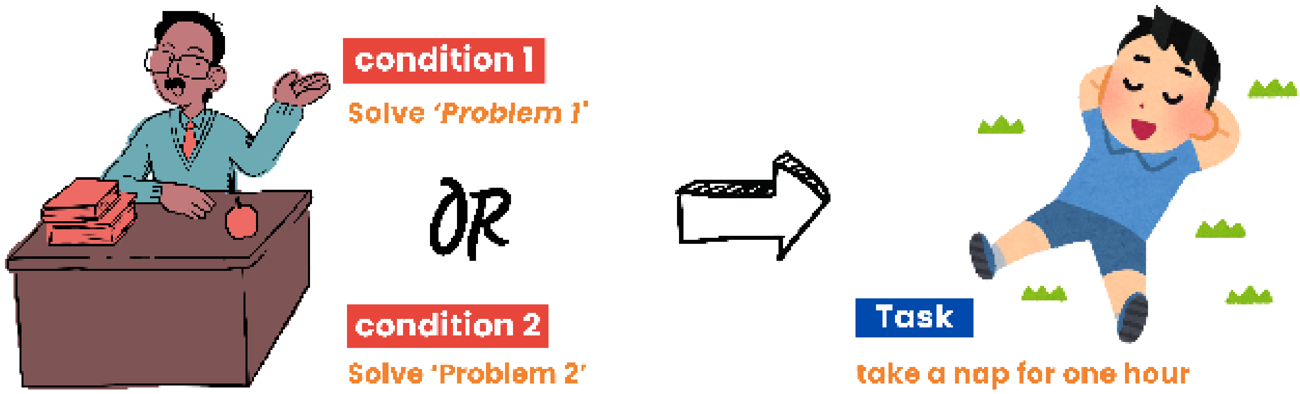
When coding, as you decide between using nested if statements or the "and" operator, consider which option will make your code clearer and more efficient for achieving the desired functionality.

Both approaches have their merits, so choose the one that best suits your specific needs!

OR

Think of the "or" operator in programming as providing options, similar to how choices might be given during school.

✓ OR



For example, imagine the principal announces during a break, "If anyone can solve either of these math problems (Problem 1 or Problem 2), then all students will get one hour of nap time today." This scenario does not require solving both problems—solving just one will grant the reward.

Similarly, in programming, using the "or" operator means if any of the conditions connected by "or" is true, the associated action will be taken.

TIPS

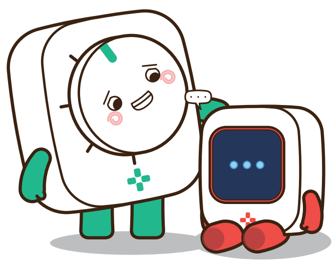
When using the "or" operator, it's crucial to manage the logical flow of your program effectively. If any connected condition is true, the overall condition is evaluated as true.

EXPLANATION WITH CODE EXAMPLE

In this example, the "or" operator offers choices. You only need to meet one condition for action to occur: **either pressing the button or detecting an object near the ToF sensor** will play the sound. If neither condition is met, the sound won't play. This flexibility allows programmers to effectively manage different scenarios, much like choosing an activity based on your situation.

Code Insight

When using the "or" operator, it's crucial to manage the logical flow of your program effectively. If any connected condition is true, the overall condition is evaluated as true.



Exercise

Exercise : Understanding "OR" and "AND" Operators

Scenario

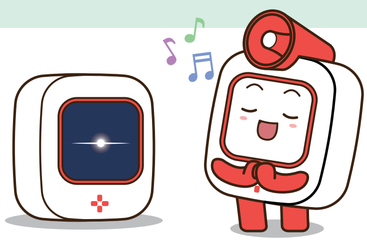
For a school science project, you've created an alarm system. The system is designed to sound an alarm if the temperature exceeds 30 degrees Celsius or if motion is detected in the room.

Question 1

The room temperature is 32 degrees Celsius and there is no motion detected. The programmer uses the "OR" operator. Does the alarm system turn on?

Question 2

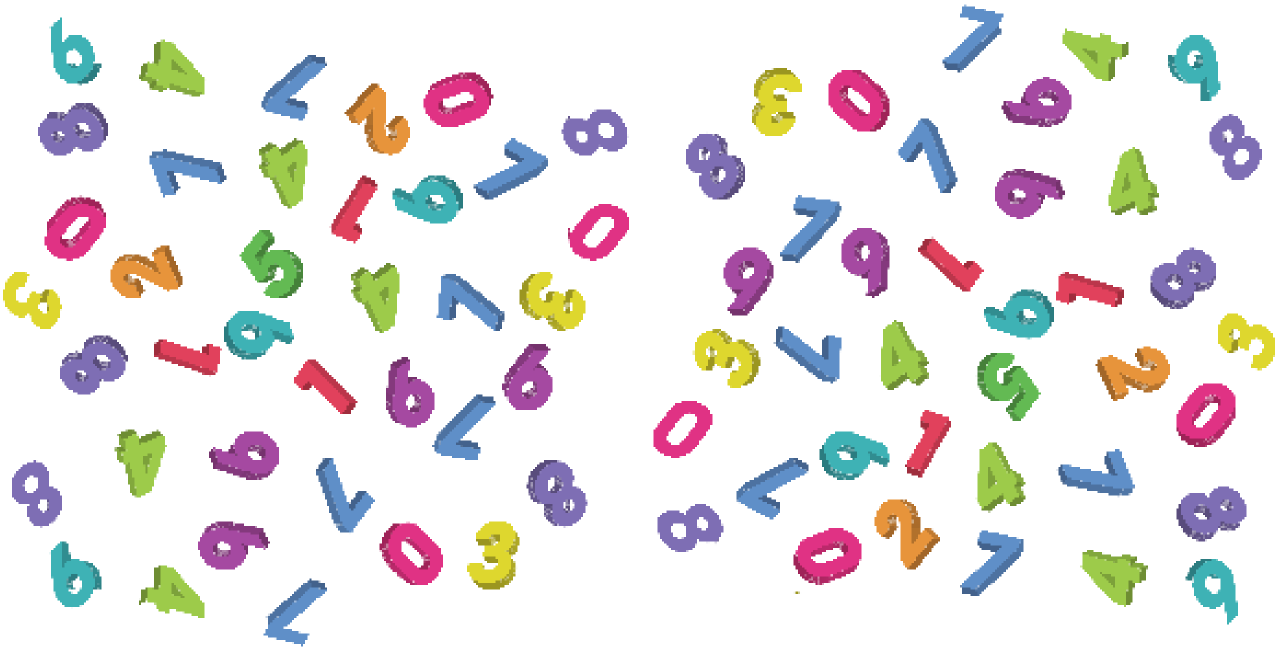
The programmer changes the conditions so that the alarm will sound only if both the temperature is above 30 degrees Celsius and motion is detected. Now, the temperature is 28 degrees Celsius, but motion is detected. Does the alarm system turn on?



Random

WHAT IS 'RANDOM'?

In programming, when we talk about something being "random," we mean it's like flipping a coin, rolling a dice, or drawing a card from a shuffled deck. You don't know what you're going to get each time. It’ s the computer's way of making a surprise choice.



HOW DOES RANDOM WORK?

Computers are very good at following instructions, but they don’ t actually make random choices the way humans can. Instead, they use special commands that create "pseudo-random" numbers. These are sequences that seem random enough for most uses even though they're actually coming from a set pattern.

WHY USE RANDOM?

1. Games

Random numbers can control things like what kind of monster pops out or what treasure you find in a video game, making the game fun and different each time.

2. Simulations

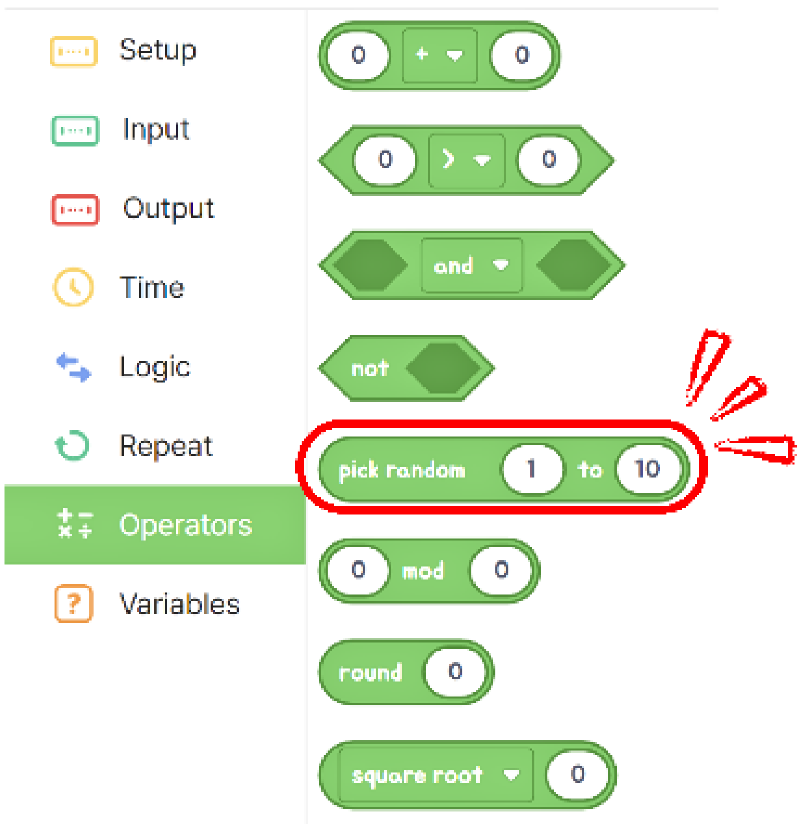
When scientists want to see how things might happen differently, like predicting weather or how traffic moves, they use random numbers to represent all the unknowns.

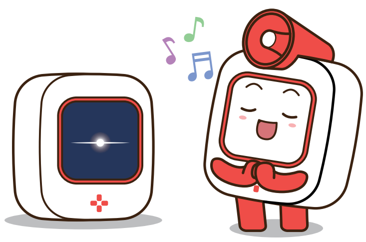
3. Safety and Security

When creating passwords or encryption keys to keep data safe, using random numbers helps make sure they're really hard for someone else to guess.

RANDOM BLOCK IN CODE EDITOR

In the operators section, there is a block for generating random numbers that allows you to specify the desired range.





Exercise

Multiple-Choice Questions

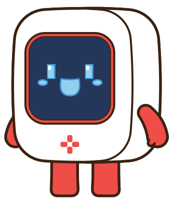
Question 1

When comparing "and" with nested if statements, what similarity do they share?

- Ⓐ Both require each condition within them to be checked sequentially.
- Ⓑ Both require all conditions to be true for the code block to execute.
- Ⓒ Both can be replaced by other logical operators in all scenarios.
- Ⓓ Both are primarily used to compare numeric values.

THINK ABOUT IT!

Do you remember?
The nested-if statement works the same as "AND"!



Question 2

How does the "or" operator differ from the "and" operator in programming logic?

- Ⓐ The "or" operator requires only one condition to be true for the code block to execute, while the "and" operator requires all conditions to be true.
- Ⓑ The "or" operator is used primarily for exclusivity checks, while the "and" is for inclusivity.
- Ⓒ The "or" operator evaluates all conditions at once, while the "and" evaluates them sequentially.
- Ⓓ The "or" operator can be used interchangeably with the "and" operator without changing the logic of the code.

THINK ABOUT IT!

What is the main difference between the "AND" and "OR"?